Activity Pages Answer Key: *Energy Transfer and Transformation*

This answer key offers guidance to help you assess your students' learning progress. Here, you will find descriptions of the expectations and correct answers for each of the Activity Pages of this unit.

Energy Scavenger Hunt (AP 1.1) (page 101)

- Students should include examples of three changes and a brief explanation of how each change occurs.
- Accept any reasonable student questions that are based on the examples the students wrote.

Surprise! (AP 2.1) (page 102)

1. Light energy transfers (moves) into the room.
2. Electrical energy transforms into light energy as the light is turned on.
3. Sound energy is made by the people talking and the noisemakers.
4. Light and heat energy are released by the candle.

Energy Causes Change (AP 2.2) (page 103)

- Students should correctly identify a change caused by the energy they circled in the word box, such as stored energy changing to motion energy when a person walks across the room.
- Before and after drawings should be correct and include the form of energy causing the change, such as drawing the candle without a flame and then a lit candle.
- Students should identify both the cause and the effect. For example, a match can be struck to light the candle, and the effect is light and heat.

On the Move (AP 3.1) (page 104)

- Students should identify an object that moves.
- Before and after drawings should be correct and show the change.
- Students' descriptions should include correct before and after descriptions based on their drawings.

Ramp It Up! (Day 1) (AP 4.1) (page 105)

Step 5 Students should complete the table for each trial. The first trial should show a lower starting height, less speed, less cup movement, and less energy. The second trial should show a higher starting height, greater speed, more cup movement, and more energy.

Step 6 Students should use their results to note the relationship between energy and speed, citing evidence to support their explanation. A steeper ramp makes the ball roll faster. The greater speed has more energy when it hits the cup, so it moves farther.

Drop It (Day 2) (AP 4.2) (page 106)

Step 6 Students should complete the table for each trial. The first trial should show a lower starting height, less speed, less impact, and less motion energy. The second trial should show a higher starting height, more speed, greater impact, and more motion energy.

Step 7 Students should use their results to describe the relationship between energy and speed, citing evidence to support their explanation. A drop from a greater height makes the ball drop faster. The greater speed has more energy when it hits, so the impact makes a bigger crater.

Lesson 6 Check (AP 6.1) (page 107)

Students should note that the pencil transfers energy of motion to the bottom of the cup and that the sound of the tapping is evidence of energy transfer. Students should also note that sound from the tapping comes out of the top of the cup and that the movement of the paper is evidence of energy transfer.

Lesson 7 Check (AP 7.1) (page 108)

1. b, c, d
2. Student drawing should show a clearly captioned and explained energy transfer by heat or electricity. The term *evidence* is used in the caption.

Investigating Energy Transfer – Station 1: Light Transfers Energy (AP 8.1) (page 109)

Step 4 Students should note that light energy is transferring from the flashlight to the wall.

1. Students should note that the flashlight batteries transfer chemical energy to light.
2. Students should note that they cannot see the beam of light without the chalk dust.
3. Students should note the chalk dust gives the light energy something to bounce off.
4. Students should note the evidence of energy transfer is the light shining on the chalk dust and wall.

Investigating Energy Transfer – Station 2: Heat Transfers Energy (AP 8.1) (page 110)

Step 3 Students should note that electrical energy is transferring to the hot plate. Then the heat energy is transferring from the hot plate to the beaker and then to the ice cubes.

1. Students should note that less heat energy would flow from the hot plate to the beaker, causing the ice cubes to melt slower.
2. Students should note that more ice cubes would take longer to melt because more heat energy would be needed.
3. Students should note the melting of the ice cubes is evidence that energy is being transferred from place to place.

Investigating Energy Transfer – Station 3: Electricity Transfers Energy (AP 8.1) (page 111)

Step 3 Students should note the electrical energy is transferring from the outlet to the surge protector and then to the device.

Station 3, continued

1. Students should note that the electrical energy flowing from the surge protector to the device causes a change.
2. Students should note that when the surge protector is off, no electrical energy flows and that when it is on, electrical energy flows.
3. Students should note that the electrical energy would not flow if there were no wire.
4. Students should note the device is signaling when electrical energy is flowing into the device.

Examples of Energy Transfer (AP 9.1) (page 112)

Students should include examples from all four types of energy listed and should include an accurate description of the transfer.

Claims, Evidence, and Reasoning About Energy Transfer (AP 9.2) (pages 113–114)

Part 1

1. Claim: A claim is a statement that declares a fact or presents an argument that has not been proven. People make claims as a key step to making a scientific examination.
2. Evidence: Evidence is proof that supports or refutes a claim. An example of evidence would be the results from a science investigation.
3. Reasoning: Student examples of reasoning should show a clear and logical progression. People use reasoning in science because they seek to reach testable truths.

Part 2

1. Student examples should indicate a clear transfer of energy, such as hot soup being poured into a bowl and subsequent temperature changes.
2. Student evidence should refer back to their learning or investigation, such as noting that the bowl that the hot soup is poured into gets hot, indicating the transfer of heat energy.
3. Student responses should be properly supported and in the form of "I know . . . because . . .," such as, "I know the heat energy in the soup transferred to the bowl because the bowl is now hot."

Lesson 10 Check (AP 10.1) (page 115)

1. The soccer player's foot and the soccer ball are about to collide.
2. The form of energy before the collision is energy of motion.
3. There would be energy of motion, which you could see as the ball moves. There would be sound energy that you could hear. There would be a tiny amount of heat energy from the collision.
4. The drumsticks and drum are about to collide.
5. The form of energy before the collision is energy of motion.
6. There would be energy of motion as the drumsticks bounce back up. There would be sound energy, which you could hear. There would be a tiny amount of heat energy from the collision.

Investigating Collisions (Day 1): Predictions (AP 11.1) (page 116)

Sample responses:

1. We want to find out how energy causes changes during a collision and if the amount of energy changes with speed.
2. We will use a ramp and send a toy car down the ramp. It will collide with a can, and we can see how far the can moves.
3. The toy car will collide with the can.
4. I predict the toy car will push the can back.
5. We will roll the car down the ramp to observe it. We will measure the distance the can moves by using car lengths.

Investigating Collisions (Day 2): Testing and Observation (AP 11.2) (pages 117–118)

3. Student observations should show that the object travelling down the ramp collided with and moved the object at the bottom of the ramp.
4. Students should identify one independent variable to change, such as the height of the ramp or the weight of an object.
5. If the toy car has more weight, it will move the can farther back when they collide.

Investigating Collisions (Day 2), continued

9.
   - Energy of motion was transferred from the toy car to the can.
   - Student observations between the two tests should be consistent with the change in variables, e.g., adding more weight to the car or height to the ramp caused a collision with more energy, resulting in the effect of the can moving farther.
   - The more energy added before the collision, the more energy we saw during the collision.
   - If more energy is added to a collision (cause), then we will see more movement (effect) after the collision.
10. We predicted adding more weight to the car would result in more energy in the collision. The prediction was accurate as the can moved farther after the collision in the second test.

Lesson 12 Check (AP 12.1) (page 119)

1. How do energy transformations help people?
2. an oven or stove
3. Electrical energy is converted to energy of motion and sound energy.
4. A kitchen timer provides the solution of what to do when you need to cook something a specific length of time so it does not overcook or undercook.
5. The stored energy of the spring inside the timer changes to energy of motion. When the bell rings, energy of motion changes to sound energy.
6. Accept all reasonable responses, such as, "The lights in the classroom allow the change of electrical energy into light energy so students can read their books."

Device Design Proposal (Day 1) (AP 13.1) (pages 120–121)

- The student problem to be solved should be their assigned problem.
- Students should identify the initial and final forms of energy their device will transform.
- Student descriptions of their device should include how the device solves the problem and some reference to how the energy will be transformed.
- Accept all plausible criteria and constraints.

(Day 1), continued

- Students should circle or list any materials they will use in their device.
- Student drawings of their device should include clear incorporation of their materials.

Device Test Results (Day 2) (AP 13.2) (page 122)

- Student results and observations may vary, but there should be improvements or fixes to their devices based on the test results.
- Claims should indicate that testing their device was an important step in improving the device.

Device Presentation Plan (Day 3) (AP 13.3) (page 123)

- Students should identify the problem, criteria, and energy transformation from Device Design Proposal (AP 13.1).
- Students should provide a clear description of how they built their device.
- Students should refer to Device Test Results (AP 13.2) for initial information about their test results, which they can then expand on.
- Students should refer to Device Test Results (AP 13.2) for initial information about their refinements to their device, which they can then expand on.

Device Presentation Scoring Guide (Day 4) (AP 13.4) (page 124)

Students should complete the scoring guide for the device. Students should also identify what they liked in the other group's device, as well as other questions they may have had about the device.

Energy Vocabulary Crossword Puzzle (AP 13.5) (pages 125–126)

ACROSS:

2. energy transfer

4. heat

8. forms of energy

11. energy of motion

12. convert

13. electricity

14. speed

15. evidence

16. change

DOWN:

1. position

3. motion

5. light

6. energy

(AP 13.5), continued

7. sound

9. stored energy

10. distance

Energy Vocabulary Review (AP 13.6) (page 127)

1. heat **2.** energy of motion, position **3.** sound **4.** transfer **5.** energy transformation **6.** change **7.** forms of energy **8.** stored energy **9.** electricity **10** speed, distance **11.** energy transformation

Energy's Big Questions (AP UR.1) (page 128)

Students should circle their assigned question. Student responses should reflect the content and experiences from the lesson(s) from which the Big Question originates. Vocabulary terms from their Core Vocabulary decks should be incorporated in their written responses and underlined.

Consider student responses as follows:

Above Average—Written responses clearly identify examples and ideas that help answer the Big Question. Examples are clearly linked to the Disciplinary Core Ideas learned, and Core Vocabulary is incorporated into the response and underlined.

Average—Written responses may include examples of learning about the Big Question but do not clearly link to the DCIs learned during the lessons. Core Vocabulary is incorporated into the response.

Adequate—The Big Question is answered, but not using examples from the lessons or Student Reader chapters. Core Vocabulary may be listed but is not used in a complete sentence, and/or a minor misunderstanding about the Disciplinary Core Ideas is present.

Inadequate—Responses do not relate to the Big Question, or responses include a clear misunderstanding of the Big Question or DCI.

Unit Assessment: Teacher Evaluation Guide

Teacher Directions: The Unit Assessment on pages 129–136 is designed as a fifty-point test. Through this assessment, students demonstrate their overall learning of the unit's Learning Objectives. CKSci Unit Assessments typically range from ten to fifteen questions in the upper elementary grades, which can be answered in a longer, single classroom session or administered in two sittings.

Items with simpler answers that assess knowledge but not the deeper understandings of the content, such as multiple choice or short answers, are weighted differently and are worth fewer points. Assessment items that require more complex thinking and a deeper understanding of the content, such as writing explanations or identifying multiple relationships, are worth more points. Items that require synthesis of content and other student knowledge are weighted with more points as well. Some test items encourage students to use their Core Vocabulary decks as a reference source for terminology and concepts related to the test item.

Expected Answers and Model Responses

1. a, b, c, d (2 points)

2. (6 points)

a) light and heat

b) light, sound (Heat and motion also should be considered correct.)

c) motion, sound

d) motion, sound (Heat should also be considered correct.)

e) motion, sound (Electrical energy should also be considered correct.)

f) electrical energy, light (Heat should also be regarded as correct.)

3. (4 points)

Above Average	Student response includes labels and/or clear descriptions of what is happening in the illustration. The type of energy causing the change is identified, and student clearly describes the change.
Average	Student response includes an accurate drawing with labels. The type of energy causing the change is identified, and student describes the change.
Adequate	Student response includes a drawing. The type of energy causing the change is identified, and/or student describes the change.
Inadequate	Student response includes an inaccurate or no drawing. The type of energy causing the change is not identified or is incorrect. Student does not describe the change.

4. Student response should note that the highest ramp will result in the greatest energy of motion when the ball contacts the block. When the energy is transferred to the block, it will move it the farthest. (3 points)

5. Student response should note that the lowest ramp will result in the least energy of motion when the ball contacts the block. When the energy is transferred to the block, it will move it the least distance. (3 points)

6. Student response should note that as the speed of an object increases, the amount of its energy of motion increases. (3 points)

7. Student response should include evidence that sound energy can travel over long or short distances. (3 points)

8. a, d, e, f, g (3 points)

9. The food lamps transfer light and heat energy to the food. (2 points)

10. Accept all plausible responses. Student responses providing evidence that the lamps transfer light and heat could include: the food stays hot on the counter, you can feel the heat coming from the lamp, and you can see the light with your eyes. (3 points)

11. Student response should note that the more weight in the car, the more stored energy it probably must have that is converted to motion energy. Based on this pattern, the car with more weight will knock more blocks over than a car with less weight. (3 points)

12. Student response should note that Student A is correct because the ball has stored energy while it is positioned in the teacher's hand. This stored energy transforms into motion energy when she drops the ball. (3 points)

13. Accept all plausible student descriptions of a device that automatically provides water and light to a plant. (4 points)

14. It could be tested by making sure the plant is healthy and the soil is moist. It might be improved by adjusting it to get more light or water. (4 points)

15. Maria's drip hose solution converts energy of position to energy of motion (as water runs down from the container to the planter). Maria's lamp solution converts electrical energy to light energy. (4 points)

Appendix A

Glossary

Blue words and phrases are Core Vocabulary terms for the unit, and Student Reader page numbers are listed in parentheses. **Bold-faced words and phrases** are additional vocabulary terms related to the unit that you should model for students during instruction and that are often used within the Student Reader, and these latter terms do not have specific page numbers listed. Vocabulary words are not intended for use in isolated drill or memorization.

C

cause and effect, n. is a relationship in which one thing makes another thing happen

change. v. to become different (1)

claim, n. a statement that answers a question or poses a solution to a problem

collide. v. to come together with impact (25)

collision, n. an instance of colliding (25)

contact, n. to be physically touching

conversion, n. the process of being converted or changing form (32)

convert. v. to change form (32)

D

design problem, n. a problem for which a design solution should be applied

design solution, n. the method someone develops to solve a design problem

distance, n. amount of space between two objects

E

electrical energy, n. the flow of electrons from one location to another

electricity, n. a form of energy caused by the movement of electrons

energy, n. the ability to cause change (1)

energy change, n. the conversion of one form of energy into one or more other forms of energy

energy of motion, n. the energy an object possesses while it is moving (9)

energy transfer, n. movement of energy from one object to another or from one place to another (17)

energy transformation, n. the change of one form of energy to another form of energy (32)

engineering design, n. a process used to develop a solution to a problem (36)

evidence, n. anything that helps prove or disprove an idea

F

forms of energy, n. the different forms energy can be found in, such as light or heat

H

heat, n. a form of energy caused by the movement of atoms or molecules

L

light, n. a form of energy caused by the movement of photons, which are little packets of energy

M

model, n. a helpful tool for representing an idea or explaining a process or relationship

motion, n. the process of an object changing position (7)

O

observation, n. the act of noting or viewing something

P

position, n. the place an object occupies

prediction, n. a statement about what might or will happen

R

reasoning, n. a scientific mindset that connects claims and evidence

rotate, v. to spin

S

solution, n. a method for solving a problem

sound, n. a form of energy that comes from a vibrating object (28)

speed, n. a measurement of the distance an object travels over an amount of time (13)

stored energy, n. energy that has the ability to cause change at a later time (4)

T

transfer, v. to move from one place to another

transform, v. to change the appearance or form of something; see also ***energy transformation***

V

variable, n. something that is changed or is not consistent

Student Safety Contract

When doing science activities, I will do the following:

- Report spills, breakages, or injuries to the teacher right away.
- Listen to the teacher for special instructions and safety directions. If I have questions, I will ask the teacher.
- Avoid eating or drinking anything during the activity unless told to by my teacher.
- Review the steps of the activity before I begin. If I have questions, I will ask the teacher.
- Wear safety goggles when working with liquids or things that can fly into my eyes.
- Be careful around electric appliances, and unplug them, just by pulling on the plug, when a teacher is supervising.
- Keep my hands dry when using tools and devices that use electricity.
- Be careful to use safety equipment like gloves or tongs when handling materials that may be hot.
- Know when a hot plate is on or off and let it cool before touching it.
- Roll or push up long sleeves, keep my hair tied back, and secure any jewelry I am wearing.
- Return unused materials to the teacher.
- Clean up my area after the activity and wash my hands.
- Treat all living things and the environment with respect.

I have read and agree to the safety rules in this contract.

_____ _____/_____/_____/

Student signature and date

_____

Print name

Dear Parent or Guardian,

During science class, we want to create and maintain a safe classroom. With this in mind, we are making sure students are aware of the expectations for their behavior while engaged in science activities. We are asking you to review the safety rules with your daughter or son and sign this contract. If you have any questions, please feel free to contact me.

_____ _____/_____/_____/

Parent or guardian signature and date